

COMP2160 GDT2 Documentation

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Features Claimed

Fill out the below table to indicate which features you have attempted for this assignment.

Feature	Weighting	Attempted? (YES/NO)
Level Maps	10%	Y
Physics-based ball movement	5%	Y
Player movement		
a) Player mesh	2%	Y
b) Directional controls	3%	Y
c) Walk / run control	2%	Y
d) Turn to face direction of movement	3%	Y
e) Remain in contact with ground at all times	3%	Y
f) Move up/down ramps	3%	Y
g) Cannot move off cliffs	3%	Y
h) Cannot move outside map	3%	Y
i) Smooth collisions	3%	Y
Kicking the ball		
a) Movement disabled while grabbing	3%	Y
b) Grab nearby ball	3%	Y
c) Control direction of kick	3%	Y
d) Trajectory line displayed	3%	Y
e) Release key to perform kick	3%	Y
Goal	5%	Y
Player animations	5%	Y
Camera		
a) 16:10 screen	2%	Y
b) Isometric camera	2%	Y
c) Move camera	2%	Y
d) Kick camera	3%	Y
e) Smooth transition between move & kick camera	3%	Y
f) Arrow pointing to off-screen ball	3%	N
User Interface		
a) Font from Font library	2%	Y
b) Start dialog	2%	Y
c) End dialog	4%	Y
d) In-game UI	2%	Y

Analytics		
a) Per session user data	4%	N
b) Per level difficulty data	3%	N
c) Per kick heatmap data	3%	N
Total	100%	

Asset Licenses

For each of the third-party assets used in your project, list the following:

- Creator (person or company name)
- Source URL (or other contact details for creator)
- License URL

KayKit: Mini-Game Variety Pack (1.2)

- Creator: Kay Lousberg
- Source: <https://kaylousberg.itch.io/kay-kit-mini-game-variety-pack>
- License: [Creative Commons Zero, CC0](#)

KayKit : Character Animations (1.2)

- Creator: Kay Lousberg
- Source: <https://kaylousberg.itch.io/kaykit-animations>
- License: [Creative Commons Zero, CC0](#)

Toony UI icons Asset Pack

- Creator: Josie Wood
- Source: <https://josie-makes-stuff.itch.io/toony-ui-icons-asset-pack>
- License: [Creative Commons Attribution-ShareAlike 4.0 International License](#)

ModularBlocks

- Creator: Chris Lee
- Source: Provided directly by creator
- License: [Creative Commons Zero, CC0](#)

COMP2160 Data Management Plan Template

(Adapted from National Archives of Australia)

1 Overview

1.1 Purpose

To determine if this game is replayable or do players only play it once, how long do they play it for?

To determine if the game is too difficult, how difficult are the levels, are there too many kicks, is the par too low?

To determine if levels are being played as designed? What area of each level do players the most time in, and where do they struggle?

1.2 Data to be measured

Question: Is the game Replayable?

Metric 1: Who is playing the game (their steam or unity user).

Metric 2: The machine the game is being played on.

Metric 3: When the player starts playing the game.

Metric 4: When the player stops playing the game.

Metric 5: Do they own similar games on steam?

Question: Is the game too hard?

Metric 1: Scene name for each level.

Metric 2: Time it takes for the level to be completed.

Metric 3: The total number of kicks that were taken to complete it.

Metric 4: Whether the player decided to retry it or move on.

Question: Are the levels being played as designed?

Metric 1: Time when the player kicks a ball.

Metric 2: The ball's position on the map.

Metric 3: The direction the ball is kicked.

1.3 Data owners and stakeholders

Position	Name	Email
Data owner <i>Has the authority to make decisions about the data</i>	Faizan E Muhammad, Samuel Constantine, Steam	47752440@students.mq.edu.au Samuel.constantine@students.mq.edu.au
Data custodian <i>Responsible for the data's technical environment</i>	Faizan E Muhammad, Samuel Constantine,	47752440@students.mq.edu.au Samuel.constantine@students.mq.edu.au
Data/information/records manager <i>Generate, access and update the data in daily activities</i>	Faizan E Muhammad, Samuel Constantine,	47752440@students.mq.edu.au Samuel.constantine@students.mq.edu.au

1.4 Related policies and regulations

- Be informed and give consent freely, specifically and unambiguously about all data collection.
- Access a copy of their data, how it is being used, who has access, and how long it will be stored for
- Have their data anonymised or pseudonymised so it cannot be tracked by unauthorised persons
- Have any inaccurate data corrected.

2 Data collection

2.1 Data description and collection methods

Provide more technical information on the data that you are collecting. This information should include:

Data to be collected	Type of data	Dataset	Collection method	Format
Who is playing the game.	Gameplay log	Player info dataset, (new set each time a player plays the game)	Using a method each time the game is played, encompassing Environment.UserName. As well as using the Steam API and Steam username.	Int as a player ID or string for the username

The machine the game is being played on.	Gameplay log	Player info dataset	Using a method each time the game is played, using Dns.GetHostEntry	Float
When the player starts playing the game.	Gameplay log	Player info dataset	Using DateTime.now in a method when the game is started	Date/Time
When the player stops playing the game.	Gameplay log	Player info dataset	Using DateTime.now in a method when the game is ended	Date/Time
Do they own similar games on steam?	Gameplay log	Player info dataset	Using the Steam API and their Steam username.	
Scene name for each level.	Gameplay log	Level info dataset - created at the completion of a level.	Using GetActive Scene	String
Time it takes for the level to be completed.	Gameplay log	Level info dataset	Using a timer which starts when the player presses start at the beginning	Float
The total number of kicks that were taken to complete it.	Gameplay log	Level info dataset	By using the already existing variable of totalKicks	Int
Whether the player decided to retry it or move on.	Gameplay log	Level info dataset	By counting when retry is pressed and when Next is pressed.	int

Time when the player kicks a ball.	Gameplay log	Kick info dataset - updated as levels are played, will be an ongoing log	Using a timer and recording when the kick happens	float
The ball's position on the map.	Gameplay log	Kick info dataset - updated as levels are played, will be an ongoing log	Using ball. position when a kick occurs	Vector 3
The direction the ball is kicked.	Gameplay log	Kick info dataset - updated as levels are played, will be an ongoing log	Using kickDirection.	Vector 3

3 Data storage and disposal

3.1 Data storage and location

Storage Location: The data collected will be stored locally on the player's computer which will then later be collected and stored locally on the game's server, The data collected from Steam also be locally stored in the Game Studios database. The data will be encrypted and will also be hidden behind multiple firewalls for data security purposes.

Data Access and Usage: The data will be accessed from the local database over time and will be utilized to fix bugs, make gameplay adjustments, change the level of difficulty for current levels and also help design upcoming levels. The data will only be accessed by authorized personnel so there are no data leaks.

Backup and Recovery: Local backups and cloud backups will be made available so that in case of an accident they can be recovered.

3.2 Data disposal (keep, destroy or transfer data)

If the user wants to delete their data or if the studio wants to dispose of the data, then all the data pertaining to that user will be deleted from the local storage and from any backups that were made of it, Also any place where the data is currently being used it will also be deleted.

3.3 Privacy

Private and Sensitive information:

- Steam name: It Can reveal some personal information of the players
- Steam ID and Names of Player's friends: It can help identify the user and their friends

Identifiable data:

- System username: It can be used to identify the person using the computer system
- IP address: it can be used to identify the location of the Player
- Steam name: Can be used to find the socials of a player as steam can be linked with other social media
- Names of Players Friends: Can be used to identify other players as well
- List of Games owned by user: Can be used to identify the preferences of the players

3.4 Ethics

Data Security: The data collected should be stored in a secure manner where it is protected from malicious attacks and leaks, the data should be stored behind several layers of encryption and firewalls and only authorized personnel should have access to the data.

Data security aligns with the ACS principles and codes, following the protection of data, ensuring privacy and managing risks. While adhering to ACS the studio should protect and safeguard the user's information.

4 Using Data

4.1 Data Analysis

The first set of data, the data relating to who is playing the game and for how long will be recorded each session that someone plays the game. This data will be used to determine whether the same people are playing the game or whether it's reaching a larger audience. It will also reveal whether people are playing the game for long periods of time or only short periods as well as whether the game is replayable or whether players play it once or twice and never again.

The second data set relates to how hard each level will be reported at the completion of each level. This data will give insight into how long it took players to complete it, whether they kept retrying and if so, how many times as well as how many kicks it took. We can compare each level across players or even the same player at different times to see whether players are completing levels faster or slower and build an accurate average of the data collected. We can compare the start and finish times for each level and see if some are taking too long or are too quick. Furthermore, can compare the number of kicks taken for each level to what the par is, this allows us to measure whether the par is too low or high for each level and therefore the game as a whole

The last set directly relates to each individual level, the data collected includes the when the player kicks the ball, where they kick it and how the direction they kick. This can be used to create heatmaps of each level which show where players spend more time kicking by comparing when and where kicks take place. This can show us if players spend more time in the parts of the level that are designed to be hard or whether they struggle with other areas of the game.

4.2 Data Implementation

The first set will be used overall as a general guide to whether the game is replayable. If the data shows players only play for a bit and don't play it again, it will encourage incentives to play the game to be introduced, maybe variations of the same levels are unlocked after completion to encourage replaying, or even tweaking pars so it's harder and more of a challenge encouraging players to keep playing.

The second set will allow a solid picture of how hard the overall game is and how long it takes to complete. If the game is completed quicker or slower than expected, future levels can be modified to be easier or harder, possibly even current levels can be made easier if the game is taking way too long to complete.

Using both the second and third sets, analysis can further improve each individual level. Having a detailed heatmap of each level as well as how long it takes to complete and how many kicks are taken will allow for developers and designers to determine whether par's need to be adjusted and whether level design needs to be changed. We can see where players struggle the most and if it's not where the level was designed to be hard, they can be tweaked to compensate for that. Designers can look at data from the current levels and change how they design future levels, making levels longer or shorter, providing different kinds of challenges or even knowing what the par should be.

Overall, the data will be analysed and used to change elements of the game's design to make it either easier or harder as well as more replayable.

4.3 Data publication

This data shouldn't be directly published or showcased anywhere and only be used for analysis. The analysis should then only be used by either developers or designers to improve the game, or marketing to market to the correct audience.

If it was to be published for some reason, however, it would need to ensure it has no specific data that can be tied back to specific players or people and generally only the analytics and statistics should be published rather than any raw data.

4.4 Data sharing

Generally, data sharing should be kept to a minimum, however, it's possible this data could be shared with similar games dev companies as well as advertisers. When doing this it's essential to ensure any personal or private data isn't shared or at least is pseudonymised so people's identity is hidden. Information such as IP addresses should not be shared as this could cause serious breaches of privacy as your player's locations could be revealed. When sharing data with external parties, only the data they need should be shared, and where possible it should be only the analytics or reports rather than raw data. These measures ensure that companies aren't accessing private data and data they don't need keeping the players safe and secure while still benefiting those companies.